

Le Thanh Tung

AI Engineer

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SKILLS

Language	Python, JavaScript, SQL
AI & Machine Learning	Machine Learning, Computer Vision, Natural Language Processing, Speech Recognition, RAG, LLMs, LangChain, LangGraph, Object Detection & Tracking
AI & Data Libraries	NumPy, Pandas, Scikit-learn, OpenCV, YOLOv8 (Ultralytics)
Backend & Databases	Flask REST API, MySQL, SQLite, MQTT, MongoDB, SQLAlchemy
Frontend Technologies	ReactJS, TailwindCSS
Extra-skill	Git

SUMMARY

- Programming Foundation:** Strong foundation in Python programming with hands-on experience developing AI applications and backend systems through academic and personal projects. Familiar with software development practices, debugging, and algorithm implementation.
- Machine Learning & Computer Vision:** Hands-on experience in implementing ML/DL algorithms, including YOLOv8 tracking/detection, OpenCV image processing, time-series forecasting, and RAG chatbots.
- Model Deployment & Full-Stack Integration:** Experienced in deploying AI models as **Flask REST APIs**, integrating AI services with databases (MongoDB, MySQL), processing multimedia using FFmpeg, and connecting backend services with web applications.
- Professional Soft Skills:** Highly proactive, self-motivated, and eager to learn new AI technologies. Strong analytical thinking, problem-solving abilities, and technical English proficiency for reading research papers and software documentation.

Extra: Actively developing personal AI projects, including speech recognition, automatic subtitle generation, sentiment analysis, music genre classification, and AI chatbot systems.

HONORS & AWARDS

2023	Won the consolation prize in English at school level
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PROJECTS

T3 - 2026 – T5 - 2026 AI Speech Recognition & Automatic Subtitle Generation

Technologies: *Python, Faster-Whisper, Flask, FFmpeg, MongoDB*

Objective:

Develop an AI-powered application capable of automatically recognizing speech from video files and generating synchronized subtitle files (SRT) with high accuracy.

Key Contributions:

- Designed an end-to-end speech recognition pipeline from video upload to subtitle generation.
- Integrated **Faster-Whisper** for speech-to-text conversion with timestamp generation and GPU acceleration.
- Developed a **Flask REST API** to process uploaded videos and return generated subtitles.
- Applied **FFmpeg** for audio extraction and multimedia preprocessing before inference.
- Implemented subtitle synchronization and formatting following the SRT standard.
- Explored LLM-based post-processing techniques to improve Vietnamese transcription quality.

Result:

- Successfully built a complete AI application capable of converting videos into synchronized subtitle files.
- Improved processing efficiency through GPU inference and optimized media preprocessing.

T1 - 2026 – T2 – 2026 Food Ordering Data Warehouse & Analytics System

Technologies: *Python, Flask, MongoDB, Pandas, ETL*

Objective:

Design and implement a data warehouse system to support sales analytics for an online food ordering platform..

Key Contributions:

- Designed ETL workflows for extracting, transforming, and loading transactional data.
- Developed backend services using Flask and integrated them with the existing web application.
- Managed order, product, and customer data using MongoDB.
- Processed analytical datasets with Pandas to support reporting and business analysis.

Result:

- Successfully integrated backend, database, and analytics components into a unified system.
- Enabled structured data analysis for sales and customer behavior.

T7 - 2025 – T9 – 2025 Music Genre Classification

Technologies: *Python, Librosa, Scikit-learn*

Objective:

Build a machine learning model capable of automatically classifying music tracks into different genres.

Key Contributions:

- Extracted MFCC and other acoustic features using Librosa.
- Developed preprocessing and feature engineering pipelines.
- Trained and evaluated supervised machine learning models.
- Compared classification performance using different evaluation metrics.

Result:

- Built an end-to-end music genre classification system.
- Demonstrated the effectiveness of MFCC-based feature extraction for audio classification.

T5 - 2025 – T7 – 2025 AI Chatbot using Retrieval-Augmented Generation

Technologies: *Python, LangChain, RAG, LLM*

Objective:

Develop an AI chatbot capable of retrieving relevant information before generating responses to improve answer accuracy.

Key Contributions:

- Implemented a Retrieval-Augmented Generation (RAG) pipeline.
- Processed and indexed documents for efficient information retrieval.
- Integrated a Large Language Model with retrieved knowledge.
- Experimented with prompt engineering techniques to improve response quality.

Result:

- Built a functional RAG chatbot prototype for document-based question answering.
- Improved response relevance by grounding generated answers on retrieved information.

EDUCATION

2022 - 2027

Information Technology

Ha Noi University Mining and Geology